Questions

Q1.

Figure 11 gives the values of gravitational field strength, *g*, on the surfaces of Earth, the Moon and Mars.

	value of *g* in N / kg
Earth	10
The Moon	1.6
Mars	3.7

Figure 11

(i) Calculate the ratio

$$\frac{\text{value of } g \text{ on Earth}}{\text{value of } g \text{ on Mars}}$$

(2)

ratio = ...

(ii) A person has a weight of 730 N on Earth.

Calculate the person's weight on the Moon, using information from Figure 11.

(2)

weight on the Moon = ... N

(iii) The gravitational field strength, g, is different on the surfaces of Earth, the Moon and Mars.

Explain why this is the case.

(2)

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(Total for question = 6 marks)

Q2.

Figure 8 shows two types of force that act in the Sun.

One type of force acts inwards, towards the centre of the Sun.

The other type of force acts outwards, away from the centre of the Sun.

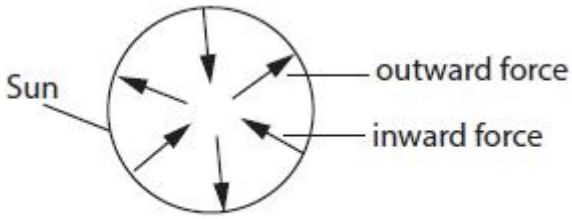

Figure 8

The inward and outward forces in stars like the Sun stay balanced for 10 billion years.

(i) In the Sun, these forces have been balanced for 46% of this time.
Calculate for how many billions of years these forces have been balanced in the Sun.

(2)

time = ... billion years

(ii) Describe what will happen to the Sun when the inward and outward forces become unbalanced.

(2)

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(Total for question = 4 marks)

Q3.

Figure 19 shows two objects, **E** and **D**.

Figure 19

E emits a sound.

D detects the sound.

E is moving in the direction shown by the arrow, but **D** is not moving.

E emits a sound of wavelength 1.86 m.

D measures the wavelength of this sound as 1.98 m.

(i) Calculate the difference between the wavelength that **E** emits and the wavelength that **D** detects.

(1)

difference in wavelength = ... m

(ii) The velocity of sound is 330 m / s.

Calculate the velocity of **E**.

(2)

Use the equation

$$\text{velocity of } \mathbf{E} = \frac{\text{velocity of sound} \times \text{difference in wavelength}}{\text{wavelength } \mathbf{E} \text{ emits}}$$

velocity of **E**= ..... m/s

(Total for question = 3 marks)

Q4.

The Sun has a mass of 2.0×10^{30} kg.
A white dwarf has a mass of 3.4×10^{29} kg.

Calculate the value of

$$\frac{\text{mass of this white dwarf}}{\text{mass of the Sun}}$$

(2)

value = ...

(Total for question = 2 marks)

Q5.

A rock on the surface of the Earth has a mass of 12 kg.

(i) Calculate the weight of this rock on the surface of the Earth.
The gravitational field strength on the surface of the Earth is 10 N/kg.
Use the equation

$$W = m \times g$$

(2)

weight on the Earth = ... N

(ii) The weight of the same rock on the surface of the Moon is 20 N.
Calculate the gravitational field strength on the surface of the Moon.

(3)

gravitational field strength on the Moon ..... N/kg

(Total for question = 5 marks)

Q6.

Figure 3 shows a Mars Exploration Rover.

(Source: *photojournal.jpl.nasa.gov*)

Figure 3

The mass of the rover is 190 kg.

(i) The gravitational field strength on Earth is 10 N / kg.

Calculate the weight of the rover on Earth.
Use the equation

weight = mass × gravitational field strength

(1)

weight on Earth = ... N

(ii) The weight of the rover on Mars is 700 N.

Calculate the gravitational field strength on Mars.

(2)

gravitational field strength on Mars = ... N/kg

(Total for question = 3 marks)

Q7.

* More than two thousand years ago the Earth was believed to be at the centre of the universe as shown in Figure 13.

The planets, the Moon and the Sun were believed to be in orbit around the Earth.

In 1543 Nicolaus Copernicus proposed that the Earth was a planet, and that the Earth and the other planets orbit around the Sun. This model is shown in Figure 14.

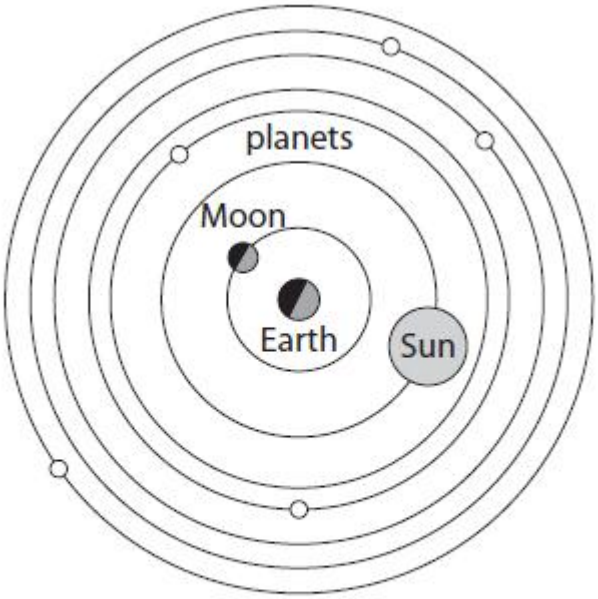

Figure 13

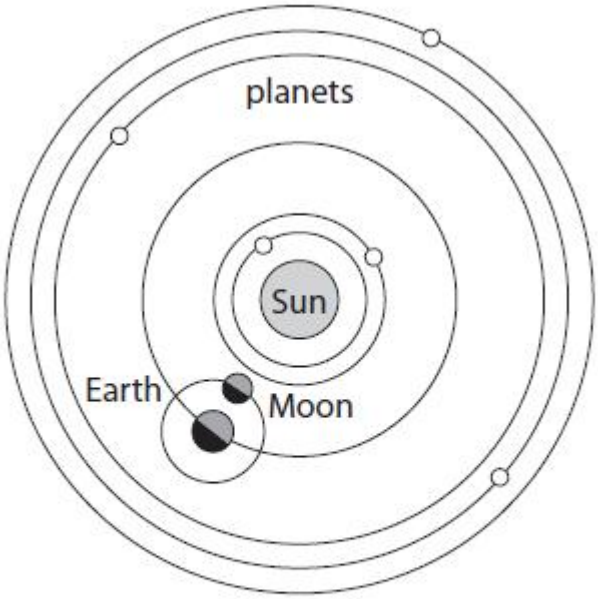

Figure 14

Discuss how evidence has changed our views of the universe.

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(Total for question = 6 marks)

Q8.

Some questions must be answered with a cross in a box(☒). If you change your mind about an answer, put a line through the box(☒)and then mark your new answer with a cross(☒).

(i) Which will be the next stage in the evolution of our Sun?

(1)

- ☐ **A** red dwarf
- ☐ **B** red giant
- ☐ **C** white dwarf
- ☐ **D** white giant

(ii) The core of a star with a much bigger mass than our Sun is most likely to end as a

(1)

- ☐ **A** white dwarf
- ☐ **B** red giant
- ☐ **C** protostar
- ☐ **D** black hole

(Total for question = 2 marks)

Q9.

The Big Bang theory and the Steady State theory are two theories about the origin of the universe.

The discovery of CMB led scientists to accept only one of the theories.

Explain why redshift supports both theories but CMB supports only one of them.

(3)

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(Total for question = 3 marks)

Q10.

A long time ago, many people thought that the objects seen in the sky were arranged as shown in Figure 17.

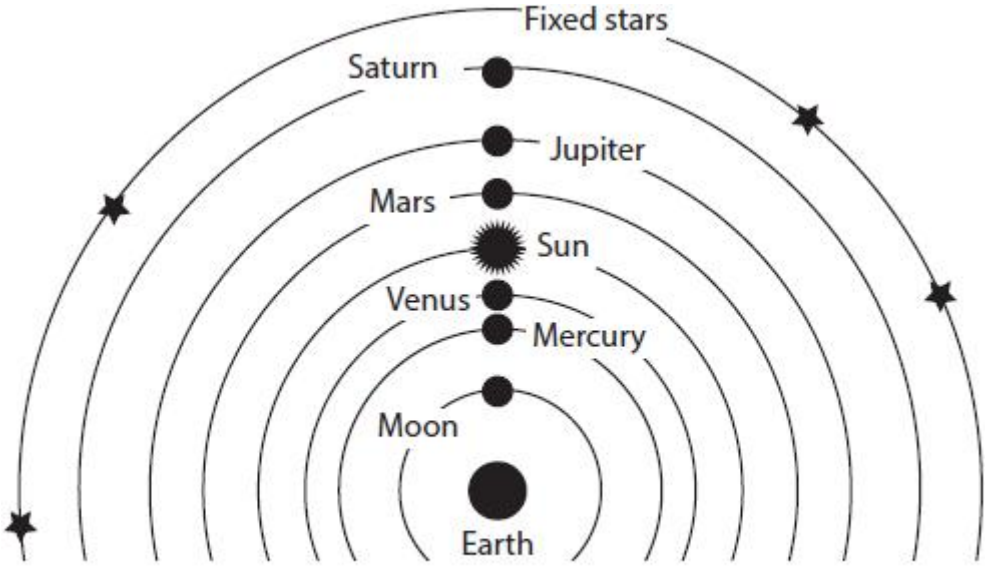

Figure 17

Compare the arrangement shown in Figure 17 with what we now know about the Solar System.

Your answer should refer to

- the objects that are in the Solar System
- how these objects are arranged.

You may draw a diagram to help your answer.

(6)

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(Total for question = 6 marks)

Q11.

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

The Steady State theory is also a theory about the origin of the Universe.

Give **one** similarity and **one** difference when comparing the Big Bang theory with the Steady State theory.

(2)

similarity

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difference

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(Total for question = 2 marks)

Mark Scheme

Q1.

Q2.

Question number	Answer	Additional guidance	Mark
(i)	$\frac{10 \times 46}{(100)}$ (1) (time=) 4.6 (billion years) (1)	accept 4 600 000 000 4.6×10^9 award one mark for power of ten error award one mark for answer of 5.4 or 5 400 000 000 or 5.4×10^9 (has found time remaining with no power of ten error) award full marks for the correct answer without working	**2** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	a description to include any **two** from {volume/ diameter/radius/ size} changes (1) (Sun becomes) a red giant (1) cools down (1) (eventually becomes) a white dwarf (1)	 ignore explode / implode do not accept red super giant / supernova lose heat (energy) accept black dwarf	**2** **AO2.1**

Q3.

	Answer	Additional guidance	Mark
(i)	(1.98-1.86) = (+/-) 0.12		**(1)** **AO2**

	Answer	Additional guidance	Mark
(ii)	(velocity =) $\frac{330 \times 0.12}{1.86}$ (1) (+/-) 21.3 (m/s) (1)	ecf from 10ai accept numbers that round to 21 (m/s) award 1,2 marks for (i) and (ii) for the correct answer for (ii) even without working	**(2)** **AO2**

Q4.

Question number	Answer	Additional guidance	Mark
	substitution (1) $\frac{3.4\ (\times 10^{29})}{2.0\ (\times 10^{30})}$ evaluation (1) 0.17	 award 1 mark for 1.7 to any incorrect power of 10 ignore any units given award full marks for the correct answer without working	**(2)**

Q5.

	Answer	Additional guidance	Mark
(i)	substitution (1) 12 x 10 evaluation (weight =) 120 (N)		**(2)** **AO2**

	Answer	Additional guidance	Mark
(ii)	rearrangement (1) $(g) = \frac{W}{m}$ substitution (1) $\frac{20}{12}$ evaluation (1) (g on Moon =)1.7 (N/kg)	rearrangement and substitution in either order accept answers that round to 1.7 e.g. 1.67 (N/kg) award full marks for the correct answer without working 1.6 scores 2 marks 240 scores 1 mark 0.6 scores 1 mark 20 × 12 scores 1 mark $\frac{12}{20}$ scores 1 mark	(3) AO3

Q6.

Question number	Answer	Additional guidance	Mark
(i)	1900 (N)	allow 1862 1864 190 × 10 190 × 9.8(1)	(1)

Question number	Answer	Additional guidance	Mark
(ii)	rearrangement (1) $\frac{700}{190}$ evaluation (1) 3.7 (N/kg)	(g =) $\frac{W}{m}$ allow numbers that round up to 3.7 (N/kg) award full marks for the correct answer without working	**(2)**

Q7.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6)** • idea that telescopic evidence supported the heliocentric model e.g. observing the moons of Jupiter supported a sun-centred solar system • detail e.g. it was Galileo's observations of the moons of Jupiter • the moons rotate around Jupiter i.e. not everything rotates about the earth • other observations by telescope – discoveries of the planets Uranus and Neptune + the asteroid belt • (detailed) photographic evidence • use of computer modelling • satellite observations • heliocentric model verified by Voyager missions (space probes)	**(6)** AO 1 1

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Q8.

	Answer	Additional guidance	Mark
(i)	**B** red giant (1)		**(1)** **AO1**

Question Number	Answer	Mark
(ii)	**D** black hole	**(1)** **AO1**

Q9.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (2 marks): • both theories predict an expanding universe and the Big Bang theory also predicts that the universe had a beginning (1) • the red shift theory indicates that the universe is expanding so supports both theories (1) • whereas CMB also indicates that the universe had a beginning, so supports Big Bang theory (1)	provided evidence that the steady state theory was incorrect	**(3)**

Q10.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. May be shown in a diagram. Objects • fixed stars are not part of solar system • additional planets are in solar system • these include Neptune and Uranus • asteroids / minor planets / comets are in solar system • planets have their own moons • the Sun is a star and is very much larger than the planets Arrangement of objects in Solar System • Earth not at the centre • Sun is in the centre • Earth between Venus and Mars • Moon (still) orbits Earth. • planets orbit the Sun. • asteroids / comets orbit Sun • orbits are not circular • (relative) distance between planets and Sun much larger • stars are very large distance from Solar System.	**6** **AO1.1** **AO3.1**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Facts about either objects or arrangement	Possible candidate responses The Earth is not at the centre. There are additional planets
Level 2	3–4	Additional guidance Partially complete comparison of objects and arrangement	Possible candidate responses Earth is a planet. Neptune is another planet. All the planets orbit the Sun.
Level 3	5–6	Additional guidance Detailed comparison of objects and arrangement	Possible candidate responses All the planets including the Earth go round the Sun. The moon goes round the Earth. The stars are a long way from the Solar System and are not part of it.

Q11.

Question number	Answer	Additional guidance	Mark
	similarity (both have) expanding Universe (1) **difference** one from: Steady State, Universe has no beginning (1) Steady State theory requires the continual formation of new matter, the Big Bang theory does not (1)	 different interpretation of CMBR	**(2)** **AO1**